ESS Technical Agreement

The Customer	GRUPA MOSTY SPÓŁKA Z OGRANICZONĄ ODPOWIEDZIALNOŚCIĄ
Project Name	Mosty Radomno Project
Project Requirement@ Nominal	1.075MW/5.015MWh
Installation Capacity	1.075MW/5.015MWh
Project Location	Radomno
Contact	Customer side: sekretariat@mosty.pl; magdalena.gradowska@mosty.pl Linyang side: kamil@linyang.com; magdalena@linyang.com.cn; kamiltyburski@linyang.com.cn
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Jiangsu Linyang Energy Storage Technology Co., Ltd

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Abbreviations

BESS	Battery Energy Storage System
BMS	Battery Management System
BMU	Battery Management Unit
PDU	Power Distribution Unit
DC	Direct Current
AC	Alternating Current
DOD	Depth of Discharge
EMS	Energy Management System
FSS	Fire Suppression System
POC	Point of Connection
SOC	State of Charge
SOH	State of Health
RMU	Ring Main Unit

1 Overview

Purpose

The purpose of this document is to clearly define LINYANG's scope of supply for Battery Energy Storage System (BESS) projects and to specify the technical characteristics of all equipment included within this scope.

Scope of Supply and Technical Features

The scope of supply and the technical specifications outlined herein are based on technical discussions and mutual understanding between the Customer and LINYANG.

Customer's Obligations

To prevent any misunderstandings or unforeseen non-conformities, the Customer shall provide LINYANG with all relevant technical requirements in writing prior to the issuance of the purchase order.

Technical Agreement

A comprehensive Technical Agreement, incorporating all technical requirements, shall be executed by both parties before the commencement of factory production. The Technical Agreement shall only be deemed valid and enforceable once it has been duly reviewed, agreed upon, and signed by both the Customer and LINYANG.

Language

All nameplates, specification sheets, and other technical information of the equipment shall be in English.

2 Standard

The battery product shall meet standard as shown below in this project.

IEC62619: Secondary cells and batteries containing alkaline or other non - acid electrolytes - Safety requirements for secondary lithium cells and batteries, for use in industrial applications

IEC63056: Secondary cells and batteries containing alkaline or other non - acid electrolytes - Safety requirements for secondary lithium cells and batteries for use in electrical energy storage systems

EN61000-6-2: Electromagnetic compatibility (EMC) - Part 6 - 2: Generic standards - Immunity standard for industrial environments

EN61000-6-4: Electromagnetic compatibility (EMC) - Part 6 - 4: Generic standards - Emission standard for industrial environments

EN62477-1: Safety requirements for power electronic converter systems and equipment - Part 1: General

The PCS product shall meet standard as shown below in this project.

IEC 62477: Safety Requirement for Power Electronic Converter System and Equipment

EN 61000: Electromagnetic Compatibility (EMC)

NC RfG: Grid Connection Requirements for Generators (Inverters, PCSs) (in compliance with EU 2016/631 and PTPiREE procedures).

3 System Design

3.1 Product Environmental Adaptability Information

No.	Project Information	
1	System Application	Outdoor
2	Temperature Range	-20~45℃
3	Humidity Range	0~95%
4	Required Anti-corrosion Level	C4
5	Altitude	<2000m
6	AC connection voltage	15kV

3.2 System Configuration

The whole solution has 1 ESS block; the total rating is 1.075MW/5.015MWh

The 1.075MW/5.015MWh block consists of 1 set of Power Conversion System (PCS)&MV Skid Unit of 1.075MW and 1 set of Battery container unit of 5.015MWh.

Each LFP battery container contains the battery system, battery management system, thermal management system, fire suppression system, and control cabinet.

Each PCS&MV Skid Unit (1.075MW) consists of 1 set of PCS (1.075MW), 1 set of 15kV transformer, and 1 set of RMU.

*Note:

1. The battery degradation of storage and shipment should be considered when testing the BESS station.
2. The test of AC usable capacity shall be conducted according to the test conditions of SAT.
3. This technical agreement does not involve the EMS part; EMS technical agreement will be provided separately upon mutual agreement by both parties, if required.
4. Auxiliary power (Battery Container: AC400V A/B/C/N/PE, 55kVA & AC230V L/N/PE 0.7kVA; MV Skid: AC400V A/B/C/N/PE, 20kVA), which is out of supply scope, shall be provided by the customer.

3.3 System Equipment List (Supply Scope)

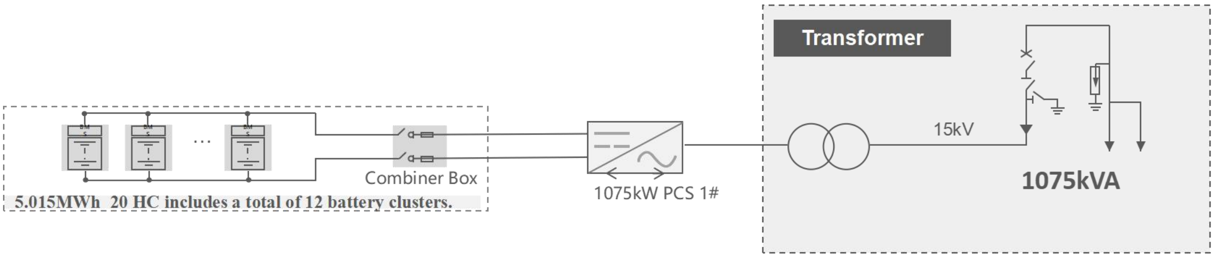

Fig. 1 Schematic diagram of a single 1.075 MW / 5.015 MWh energy storage unit (Reference)

Tab. 1 System Equipment List (Supply Scope)

1 MW / 5.015 MWh BESS Equipment Configuration List					**Location：Poland**
No.	**Item**	**Model & Specification**	**Unit**	**Qty.**	**Remark**
1	**Battery container**	**20ft 5.015MWh LFP battery system**	**Set**	**1**	
1.1	Battery	5.015MWh (314Ah, 1P104S4S*12P, 0.5P, battery packs, control cabinet, battery strings & system accessories)	Set	1	- EVE battery cell
1.2	BMS	Including BMU、BCU、BAU three-level architecture	Set	1	
1.3	Control cabinet	12 in 1	Set	1	
1.4	Thermal management System	Liquid cooling system	Set	1	
1.5	Fire Suppression system	Combustible gas detection, ventilation system, Aerosol fire suppression, water fire suppression system	Set	1	
1.6	Enclosure and Aux.	Dimension: (W*H*D) 6058×2896×2438mm	Unit	1	- C4 - IP55 rating - ≤2000m - Auxiliary power provided by customer: AC400V A/B/C/N/PE, 55kVA & AC230V L\N\PE 0.7kVA - Paint Colour: Door & Container body RAL9003; Bottom RAL9005
2	**PCS&MV Skid**	**1.075MW**	**Set**	**1**	
2.1	PCS	PWS1-1725KTL-H-EX-5M1-O	Set	1	- SINEXCEL

		AC output power:1075 kVA@45℃ Nominal AC voltage: 690V			- C4 - IP65 rating
2.2	Step-up Transformer	1075kVA/ 15-0.69kV, oil immersed	Set	1	-1075kVA@45℃ - Dy11 - Al/Al - Tier 2 - Transformer body: IP68, Other parts: IP54 - Mineral oil
2.3	RMU	24kV	Set	1	- MINGYANG -SF6-free - Indoor design - 630A
2.4	Enclosure and Aux.	Dimensions: (W*H*D) 6058×2896×2438mm	Pcs	1	- C4 - IP54 rating - ≤2000m - Auxiliary power provided by customer: AC400V A/B/C/N/PE, 20kVA - Paint Colour: Door & Container body RAL9003; Bottom RAL9005

Note:

1. The cable supply scope contains internal cables only. Power and communication cables between equipment are not included in the supply scope.

2. The rated energy storage capacity at BOL is 1.075MW/5.015MWh

3. Product applicable altitude: <2000m; Ambient temperature range: -20℃~45℃; Anti-corrosion grade: C4, Protection grade: IP55 (battery container), IP54 (MV Skid).

4. Grid support: Active & reactive power control, grid following.

4 System Introduction

4.1 Battery Unit

Fig. 2 Appearance of 5.015MWh container (Reference)

Tab. 2 Parameters of energy storage system

Item		Specification
Battery type		314Ah
Configuration		12P416S
Rated energy/ kWh		5015
Battery voltage range/ V		1164.8 ~ 1497.6
Duration/ h		≥2
Communication		Ethernet / CAN / RS485
Thermal Management		Liquid cooling
Fire Suppression System		Automatic fire protection system & explosion-proof system & emergency water system
Dimension (W x D x H)/ mm		6,058 x 2,438 x 2,896
Weight/ T		~ 41.5
Operation temperature/ ℃	Charge	0 ~ 50
	Discharge	-20 ~ 50
Humidity (storage/operating)		＜95% RH (non-condensing)

Max. working altitude	2000m
IP level	IP55
Degree of anti-corrosion	C4
Communication interfaces	Ethernet
Certificates	IEC62619, IEC 63056, IEC 62477-1, IEC61000, UN38.3/UN3536

4.1.1 Battery Cell

Fig. 3 Appearance of battery cell (Reference)

Tab. 3 Specification of battery cell

Item		Specification
Battery type		LFP
Capacity/ Ah		314
Nominal voltage/ V		3.2
Voltage range/ V		2.5 ~ 3.65
Rated energy/ Wh		1004.8
Charge/Discharge Current	Standard	0.5P
Dimension (W x D x H)/ mm		(207.7±1.0) x (71.7±2.0) x (174±1.0)
Weight/ Kg		5.6±0.3
Operation temperature/ ℃	Charge	0 ~ 55
	Discharge	-30 ~ 55
Certificates (All finished)		UL 1973 / UL 9540A / IEC 62619 UN 38.3 / GB/T 36276 / ROHS / Bankability

4.1.2 Battery Pack

Tab. 4 Parameters of battery pack

Item		Specification
Pack type		1P104S
Capacity/ Ah		314
Rated energy/ kWh		104.499
Configuration		1P104S
Nominal voltage/ V		332.8
Voltage range/ V		260-397.6
Duration/ h		⩾2
Operation temperature/ ℃	Charge	0 ~ 50
	Discharge	-20 ~ 50
Dimension (W x D x H)/ mm		(790±5) *(2180±5) *(252±5)
Weight/ Kg		~690±10
IP level		IP67
Thermal management		Liquid cooling
Certificates		IEC 62619/63056/EN IEC 61000-6-2/61000-6-4

Fig. 4 The diagram of battery pack (Reference)

4.1.3 Battery String

Tab. 5 Parameters of battery string

Item	Specification
String type	1P416S

Battery type		LFP
Capacity/ Ah		314
Key Component		4 Packs
Rated energy/ kWh		417.996
Configuration		1P416S
Nominal voltage/ V		1331.2
Voltage range/ V		1164.8 ~ 1497.6
Duration/ h		≥2
Operation temperature/ ℃	Charge	0 ~ 50
	Discharge	-30 ~ 50
Thermal management		Liquid cooling
Certificates (All finished)		IEC 62619/63056/EN IEC 61000-6-2/61000-6-4

4.1.4 Battery Management System (BMS)

Tab. 6 Technical parameters and indicators required for the battery management system

Level	Configuration	Functionality
Level 1	BMU *1 per Pack	1. Detects individual cell voltage and temperature, compiles the collected data, and reports it. 2. Acts as the balancing unit for cell voltage, performing autonomous balancing or following commands from higher-level units. 3. Conducts real-time self-diagnosis of sampling and balancing circuits, reporting results in real-time. 4. Communicates with the higher-level BCMS through Daisy Chain communication.
Level 2	BCMS *1 per battery string	1. Collects temperature, voltage, current, and SOC information uploaded by battery packs within the string. 2. Calculates battery SOH based on calibration operations. 3. Manages the normal operation and alarm information of the battery string, controlling the opening and closing of the string's DC switch as needed. 4. Issues balancing commands based on the voltage status within the string. Issues insulation detection commands as per higher-level instructions. 5. Uploads battery string parameters and switch status to the BAMS through Ethernet or CAN communication, and receives control instructions from BAMS.
Level 3	BAMS *1	1. Collects and manages BCMS information from the second level through the switch, overseeing voltage,

		current, temperature, and SOC of lower levels. 2. Manages the direction of the energy flow of the storage system in coordination with the EMS, SCADA, and PCS management systems. 3. Monitors bus voltage, total current, and bus insulation. 4. Consolidates overall alarm and protection statuses and decides whether to control the main circuit breaker for protection. 5. Calculates the SOC of connected batteries and calibrates battery capacity based on the cycle status. 6. Transmits data to the touchscreen and responds to local control commands issued from the touchscreen.

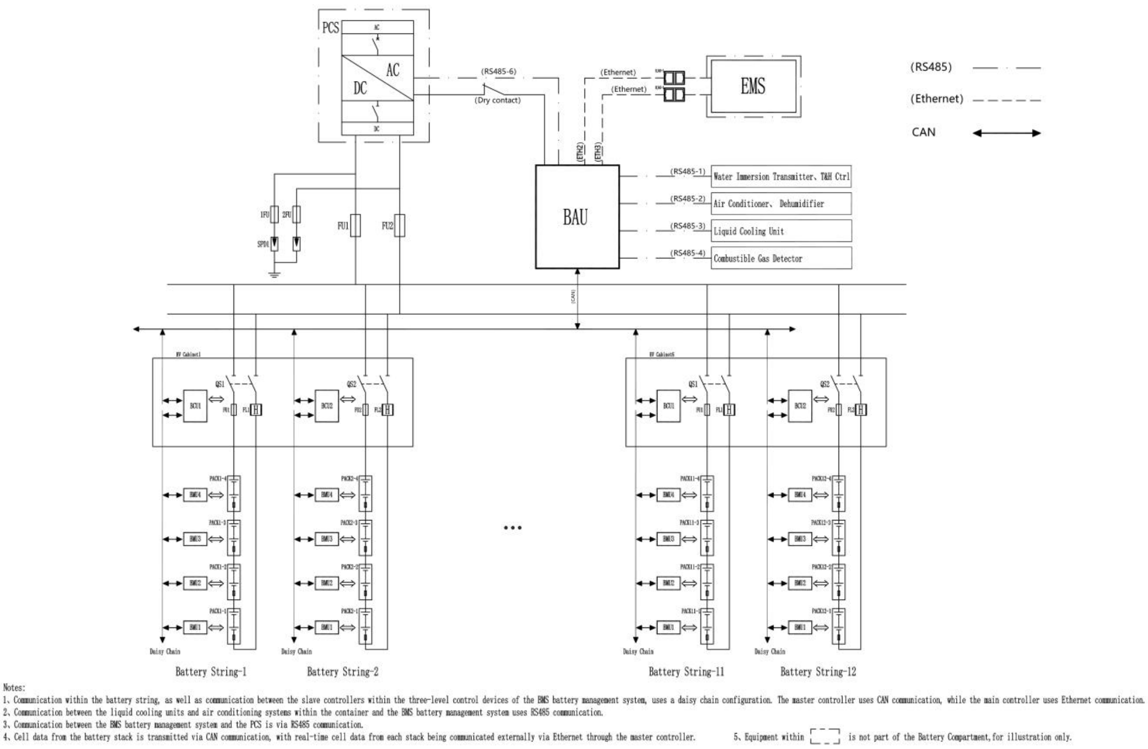

Fig. 5 Communication framework of BMS (Reference)

Tab. 7 Fault diagnosis item list

No.	Alarms and Fault Protection
1	Cell Charging Voltage Primary Warning
2	Cell Charging Voltage Secondary Warning
3	Cell Charging Voltage Tertiary Warning
4	Cell Discharging Voltage Primary Warning
5	Cell Discharging Voltage Secondary Warning
6	Cell Discharging Voltage Tertiary Warning
7	Pack Charging Voltage Primary Warning

8	Pack Charging Voltage Secondary Warning
9	Pack Charging Voltage Tertiary Warning
10	Battery String Charging Voltage Primary Warning
11	Battery String Charging Voltage Secondary Warning
12	Battery String Charging Voltage Tertiary Warning
13	Battery String Discharging Voltage Primary Warning
14	Battery String Discharging Voltage Secondary Warning
15	Battery String Discharging Voltage Tertiary Warning
16	Level 1 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Charging
17	Level 2 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Charging
18	Level 3 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Charging
19	Level 1 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Discharging
20	Level 2 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Discharging
21	Level 3 Alarm Value for Voltage Range Difference of Battery Cells in Battery String During Discharging
22	Battery String Charging Current Primary Warning
23	Battery String Charging Current Secondary Warning
24	Battery String Charging Current Tertiary Warning
25	Battery String Discharging Current Primary Warning
26	Battery String Discharging Current Secondary Warning
27	Battery String Discharging Current Tertiary Warning
28	Cell Overtemperature Primary Warning
29	Cell Overtemperature Secondary Warning
30	Cell Overtemperature Tertiary Warning
31	Cell Lowtemperature Primary Warning
32	Cell Lowtemperature Secondary Warning
33	Cell Lowtemperature Tertiary Warning
34	Level 1 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Charging
35	Level 2 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Charging
36	Level 3 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Charging
37	Level 1 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Discharging
38	Level 2 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Discharging

39	Level 3 Alarm Value for Temperature Range Difference of Battery Cells in Battery String During Discharging
40	Cut-off Threshold of Cell Temperature Maximum Deviation during Battery String Charging
41	Cut-off Threshold of Cell Temperature Maximum Deviation during Battery String Discharging
42	Level 3 Insulation Resistance Alarm Signal of Battery String
43	Level 1 Insulation Resistance Alarm Signal of Battery String

Tab. 8 Data upload list（by the final design）

No.	Data Type	Data Name	Remarks
1	Battery stack status	Alarm Total	
2		Fault Total	
3		Current Status	Idle, Charging, Discharging
4	Battery Stack information	Battery Stack Average SOC	
5		Battery Stack Discharge Power Limit	
6		Battery Stack Charge Power Limit	
7		Battery Stack Discharge Current Limit	
8		Battery Stack Charge Current Limit	
9		Current Total Discharge Capacity	
10		Current Total Charge Capacity	
11		Battery Stack Voltage	
12		Battery Stack Current	
13		Battery String Operating Status	
14		Battery String Charge/Discharge Status	
15		Battery String Operating Mode	
16		Battery String Control Mode	
17	Battery String information	Battery String Total Current	
18		Battery String Operating Power	
19		Battery String Total Voltage	
20		Battery String Overall SOC	
21		Battery String Operating SOC	
22		Battery String Available SOC	
23		Battery String SOH	
24		Battery String Charge Current Limit	
25		Battery String Discharge Current Limit	
26		Battery String Charge Power Limit	
27		Battery String Discharge Power Limit	
28		Battery String Charge Capacity	

No.	Data Type	Data Name	Remarks
29		Battery String Discharge Capacity	
30		Battery String Minimum Cell Voltage	
31		Battery String Minimum Voltage Cell Group Number	
32		Battery String Minimum Voltage Cell Number	
33		Battery String Maximum Cell Voltage	
34		Battery String Maximum Voltage Cell Group Number	
35		Battery String Maximum Voltage Cell Number	
36		Battery String Minimum Cell Temperature	
37		Battery String Minimum Temperature Cell Group Number	
38		Battery String Minimum Temperature Cell Number	
39		Battery String Maximum Cell Temperature	
40		Battery String Maximum Temperature Cell Group Number	
41		Battery String Maximum Temperature Cell Number	
42	Cell information	Cell Voltage	
43		Cell Temperature	

Tab. 9 BMS electrical parameters

Item		Parameter	Remarks
*Balancing Method		Passive	
*Balancing Current		⩾100mA	
*Balancing Efficiency		/	
*Power Consumption	Cell Monitoring Module Power Consumption	⩽2W	
	BMS Control Module Power Consumption	⩽3W	
	BMS Display and Control Module Power Consumption	⩽15W	

Tab. 10 BMS data collection parameters

Item		Parameter	Remarks
Voltage collection	Group Terminal Voltage Detection Range	<1500V	
	Group Terminal Voltage Detection Accuracy	±0.5%	
	Single Cell Voltage	1.5~4.5V	

	Detection Range		
	Single Cell Voltage Detection Accuracy	±5mV	
	Voltage Collection Interval	≤100ms	
	Voltage Transmission Mode	Change-based Upload+ Periodic Upload	
	Voltage Transmission Interval	≤5s	
Current collection	Current Detection Range	±300A	
	Current Detection Accuracy	≤±0.5%	
	Current Collection Interval	≤50ms	
	Current Transmission Mode	Change-Based Transmission + Periodic Transmission	
	Current Transmission Interval	≤1s	
Temperature Collection	Temperature Detection Range	-40～125℃	
	Temperature Measurement Accuracy	±1℃	
	Temperature Resolution	≤1℃	
	Temperature Collection Interval	≤1s	
	Temperature Transmission Mode	Change-Based Transmission + Periodic Transmission	
	Temperature Transmission Interval	≤5s	
Isolation Resistance Detection	Detection Accuracy	<10%	

Tab. 11　BMS computation & storage parameters

Item		**Parameter**	**Remarks**
*SOE Calculation Accuracy		≤5%	Self-calibration function recommended
SOE Calculation Update Cycle		≤3s	
Energy Calculation Error		≤±3%	
Event Logging		≥100000 lines	Accurate to sec
Information Storage Duration		Stored online >180 days	
Fault	Current	≤50ms	

Recording	Recording Interval		
	Voltage Recording Interval	≤1s	
	Temperature Recording Interval	≤5s	
	Recording Duration	≥10min	

Tab. 12 BMS other parameters

Item	Parameter	Remarks
BMS System Architecture	Three-tier BMS Architecture	
Protection Level	IP 3X	Dust Protection Measures Required
*Durability	10years	

4.1.5 Liquid Cooling System

The battery part of the system adopts liquid cooling technology to dissipate heat. Liquid cooling technology brings less loss and better temperature uniformity. The liquid cooling system mainly comprises of pipes, pumps, heat exchangers and compressors. The coolant of the system is mixed solution of ethylene glycol and water. The coolant flows from the water inlet pipe to the 6 main longitudinal branches. Each main branch is divided into 8 branches and flows to the liquid cooling pack to cool / heat the cell. After flowing out of the pack, it is summarized to the return pipe through the branch pipe and returned to the liquid cooling unit.

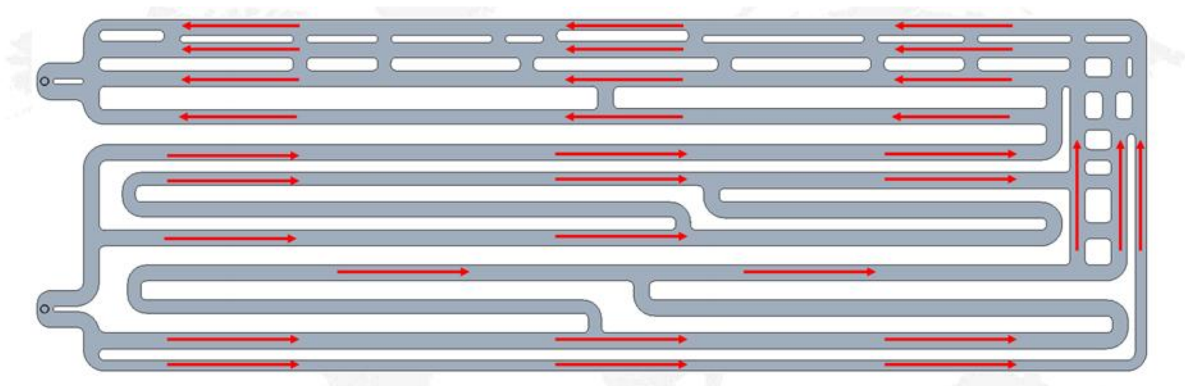

Fig. 6 Flow direction of coolant in pack (Reference)

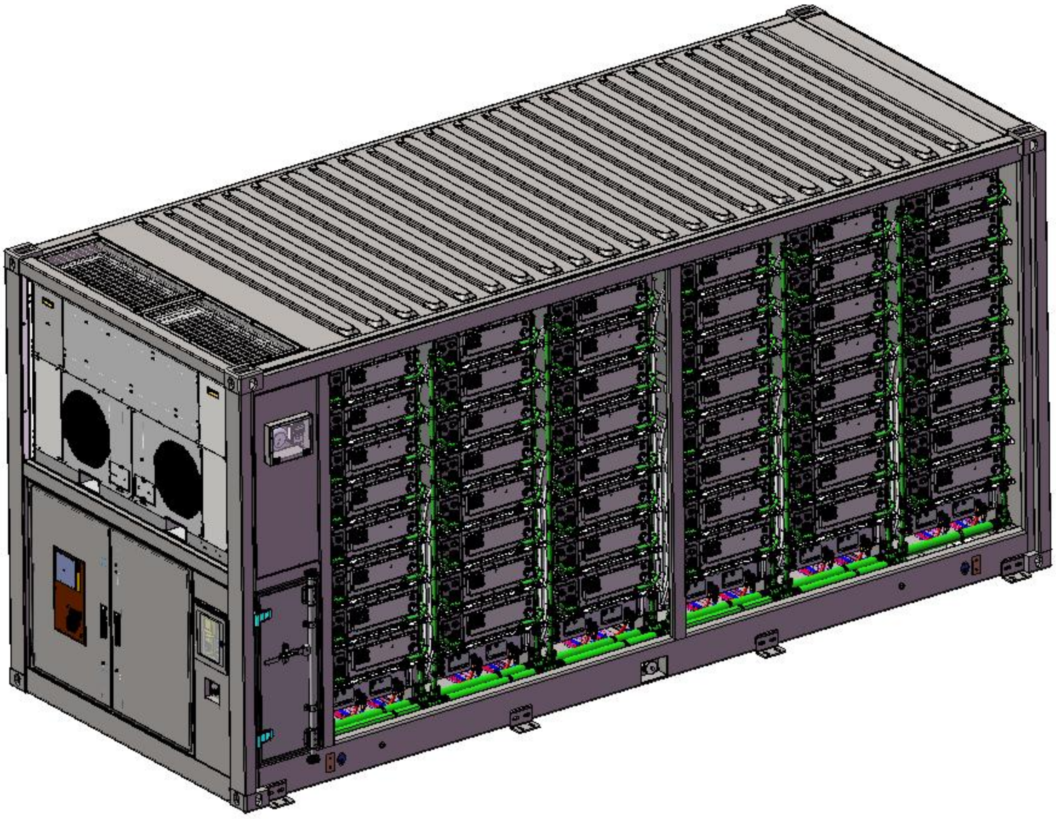

Fig. 7 Liquid cooling design of system (Reference)

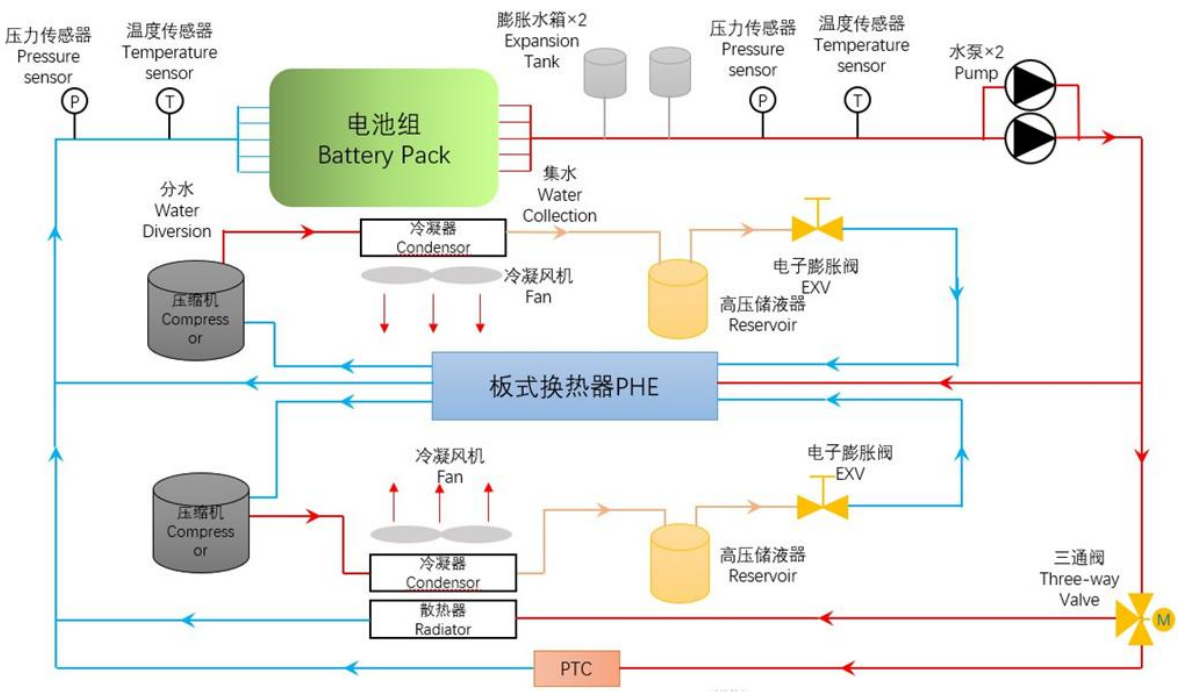

Fig. 8 System schematic (Reference)

Tab. 13 The liquid cooling unit main technical parameters

Model	BTMS-600-ESB11

Cooling Capacity / kW	60(The ambient temperature is 45 °C and the outlet water temperature is 20 °C)
Heating Capacity / kW	16
Total Power Supply / V	480V (323~552V)
Frequency / Hz	50/60
Phase	3
No. of Wires	5
Total A – Cooling / A	43
Maximum Operating Current / A	60
Total A – Heating / A	26
Refrigerant, Type	R513a
Amount / oz	113*2
Design Pressure	-
High Side / psig (MPa)	319 (2.2)
Low Side / psig (MPa)	217 (1.5)
Operating Ambient Temperature / ℃	-30~55
Coolant Outlet Temperature / ℃	5~35(Cooling); ＜40(Heating)
Rate Of Flow / L/Min	500(@150kPa
Altitude / m	0~4000
IP	IP26(Entire machine) IP67(Electrical components)
Cabinet Outer Enclosure	
Length / mm (max)	2100
Depth / mm	550
Height / mm (max)	1230
Thickness / mm (min)	2.5
Material	Aluminum
Weight / kg	≤410
Color	RAL9003

4.1.6 Fire Suppression System

The designs the container fire protection solution with automatic fire protection system + explosion-proof system + emergency water system (see topology and logic diagram of the fire protection system of the container of electrochemical energy storage), which strictly follows the relevant national standards and norms in order to ensure the scientificity, reasonableness and validity of the scheme, comprehensively safeguard fire protection of the electrochemical energy storage system, and ensure that the system can be safely run and highly effective response to fire accidents.

(1) Ventilation systems:

When combustible gas is generated from thermal runaway of the battery cell in the container, the combustible gas detector will immediately activate the ventilation system to ventilate the area when it detects that the concentration of combustible gas has reached a preset value, so as to prevent the risk of explosion when the combustible gas reaches the explosive concentration and encounters external energy.

(2) Automatic fire-fighting systems:

Each battery pack integrates one unit of QRR0.144G/S condensed-aerosol fire-suppression module, monitored by an embedded temperature-sensing wire. When thermal runaway drives the cell temperature to the preset threshold, the wire immediately registers the event and actuates the module, achieving rapid knock-down and preventing fire propagation.

When a fire occurs in the container, the smoke detector and the temperature detector detect the fire and transmits the signal to the gas fire suppression controller, which activates the aerosol fire extinguishing device in the container to start the fire extinguishing.

(3) Emergency water system:

Due to environmental or other factors, the automatic fire suppression system fails or the fire spreads to become a fire, professionals need to connect the water from the municipal fire hydrant to the water fire interface on the container to start the emergency water supply system.

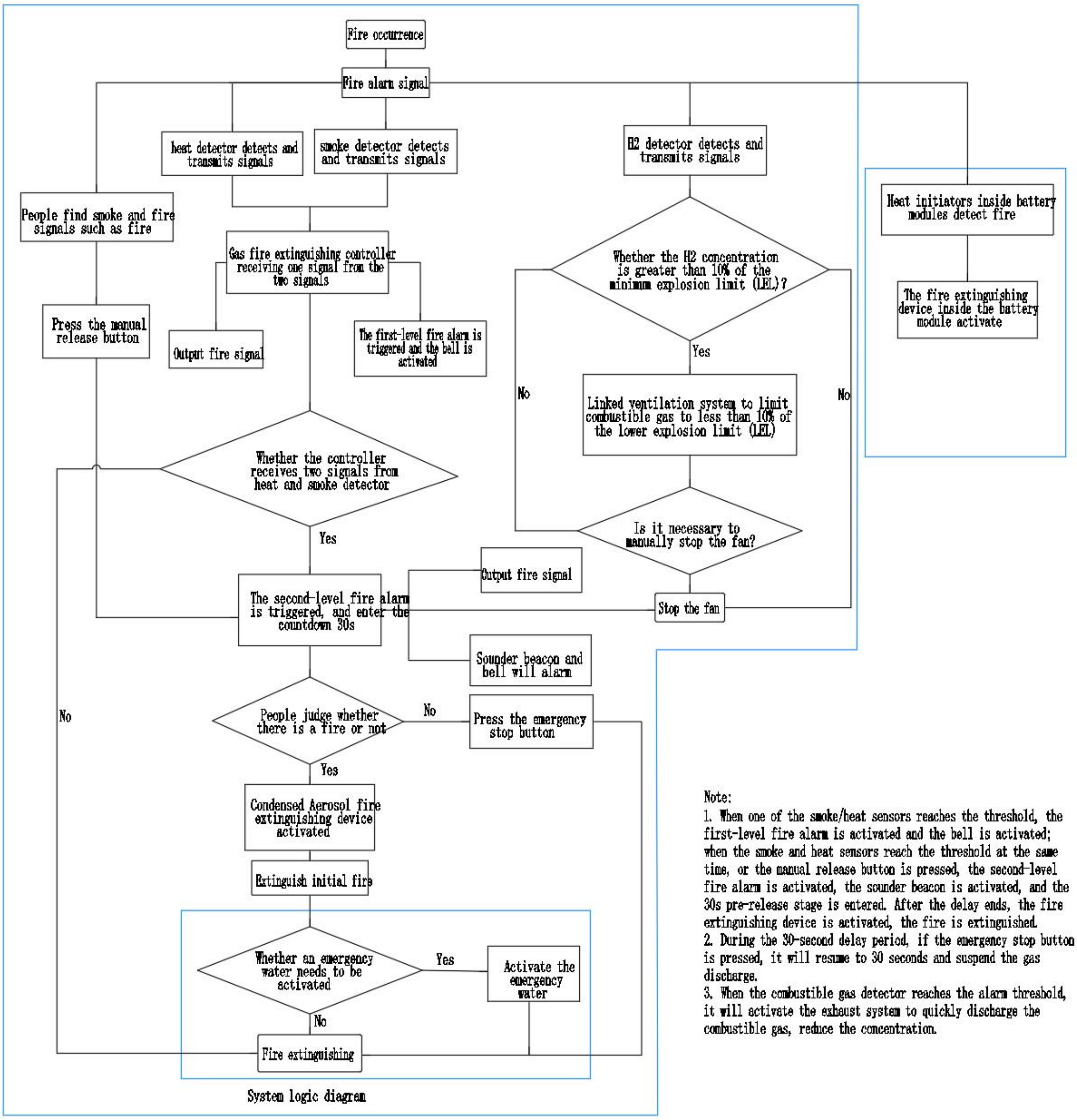

Fig. 9 Logic diagram of the container BESS and fire protection system (Reference)

Tab. 14 The components in fire suppression system

No	Protective zone	Name	Model	Unit	Qty	Remark
1	Automatic fire fighting systems	module level condensed aerosol fire extinguishing	QRR0.144G/S-MS-144-F-02-11	Pcs	96	
2		Condensed aerosol fire extinguishing device	QRR0.75G-U01	Pcs	5	

3		Extinguishant control panel	K11031M2	Pcs	1	
4		Smoke detector	55000-316	Pcs	2	
5		Heat detector	55000-121	Pcs	2	
6		Detector base	45681-200APO	Pcs	4	
7		Sounder beacon	958CHL1000	Pcs	1	
8		Bell	CBE6-RS-024-EN	Pcs	1	
9		Extinguishing hold off	K91000M10	Pcs	1	
10		Extinguishant Status Indicator	K911110M8	Pcs	1	
11		Backup battery	SSP12-7	Pcs	2	
12	Explosion proof exhaust system	Air intake fan	CNBF-350A	Pcs	1	
13		Explosion-proof exhaust fan	CNF-350A	Pcs	1	
14		Fan star & stop button	/	Pcs	1	
15		Combustible gas detector	Xgard Bright-H_2	Pcs	2	
16	Emergency water system	/	/	SET	1	
17	Auxiliary materials	/	/	SET	1	

4.2 Auxiliary Power

The project relies on an external power supply, and the Owner shall provide such external power to serve as the auxiliary power for equipment in the Battery container and MV Skid. The specific required parameters are as follows:

Supply Type	Power Supply	Voltage Level	Frequency	Recommended Cable (Reference)
External Auxiliary Power Supply for Battery container	55kVA	AC400V A/B/C/N/PE	50Hz	4*2AWG+1*5AWG
	0.7kVA	AC230V L\N\PE	50Hz	3*8AWG

External Auxiliary Power Supply for MV Skid	20kVA	AC400V A/B/C/N/PE	50Hz	4*3AWG+1*5AWG

4.3 PCS&MV Skid

4.3.1 Product Specification

Tab. 15 Technical Parameters of PCS&MV Skid

Technical Specification	
Items	**PWS1-1725KTL-H-EX-5M1-O**
DC input	
Max. DC voltage	1500Vdc
DC voltage range	1000-1500Vdc
Max. DC current	1005A
Soft start	Yes
AC output (On-grid)	
Rated AC output power	1075kW
Max. AC output power	1183kVA
Rated grid-tied voltage	690Vac
Grid voltage range	-15%~10% (settable)
Grid frequency range	50Hz
Max. output current	989A@690V
Power factor	>0.99 (at rated power)
Adjustable power factor	1 (leading)~1 (lagging)
THDi	<3% (at rated power)
AC output (Off-grid)	
Rated AC output voltage	690Vac
Output voltage accuracy	1%
Max. output current	989A@690V
THDu	<3% (liner load)

Rated output frequency	50/60Hz
PCS efficiency	
Max. efficiency	98.5%
Transformer	
Rated power	1075kVA
Voltage transformation ratio	0.69/15kV
Isolation mode	oil-immersed transformer (without emergency oil tank)
General data	
IP rating	IP54
Operation temperature	-20℃~60℃ (>45℃ derating)
Relative humidity	0~100% (no-condensing)
Cooling type	Intelligent air cooling
Dimensions (W×H×D)	6058×2896×2438mm
Weight	<15000kg
Altitude	2000m (>1000m derating)
Display	LED
Communication protocol	Modbus-RTU, Modbus-TCP, IEC61850, IEC104
Communication interface	RS485, CAN, Ethernet

NOTES:

1. The PCS can work at all 4 quadrants with unity leading and lagging power factor.

4.4 RMU

The high-voltage switchgear serves as a critical device for connecting and interrupting the high-voltage side of the transformer with the upper-level power system. It includes necessary electrical equipment such as high-voltage vacuum circuit breakers, high-voltage earthing switches, high-voltage bushings, surge arresters, live line indicators.

Items	Specifications
Brand	MINGYANG
Switchgear Type	Ring Main Unit
Rated voltage (kV)	24
Insulating medium	SF_6-Free
Rated frequency (Hz)	50
Enclosure protection degree	IP4X
Gas tank protection degree	IP67
Rated Operating Current (A)	630
Switchgear Short Circuit Rating (kA/s)	25/4s

5 Disclaimer

(1) The LINYANG shall not be held liable for defects, damages or underperformance of the device caused by the following conditions during the warranty period.

- The on-site storage time of the device exceeds the limit specified in the contract.
- The external ambient temperature and humidity for storage/operation of the device are beyond the allowable range.
- The Customer fails to operate or maintain the device in accordance with Product Specifications and System Manual.
- The device is used beyond the specified scope in the technical agreement.
- The Customer fails to charge and discharge the battery within the storage limit in accordance with the requirements of the LINYANG. The device shall be stored for no more than 6 months after it leaves the LINYANG factories.
- Allowing any battery string to hold a 0% SOC condition or allowing any cell voltage to be held at or below 2.7V (Minimum Cell Voltage) for more than 120 consecutive hours during maintenance or shutdown period.
- Force Majeure, including government acts, wars, and natural disasters.
- Other faults or defects that are not caused by the LINYANG.

(2) Adjustments shall be made to the capacity performance guarantee specified in the agreement, where capacity losses occur under the following three conditions:

- The Customer caused delay in commissioning and grid connection.
- Due to reasons attributable to the Customer, the storage temperature has deviated

from the specified range.

- Failure of the Customer to set RSOC of the ESS in accordance with the specified range during usage.

(3) The capacity mentioned in this chapter shall be tested in accordance with the requirements of the LINYANG.

(4) The LINYANG only provides the equipment, and does not provide any primary or secondary cables outside the container or between container. The LINYANG provides remote guidance wiring, but is not responsible for the laying, wiring, plugging, etc. of the cables.

(5) If the energy storage container is not put into operation or connected to the grid, it needs to be regularly charged (at least once every 3 months) If a series of problems arise due to the battery being discharged due to long-term storage, LINYANG shall not be responsible.

(6) The energy storage container is a standard product. If it is used in a specific project and involves functions other than the standard functions, communication with LINYANG is. If the standard functions cannot meet the needs, customization development will be necessary, and there will be development costs.

6 Signature

Signed by

Signed by